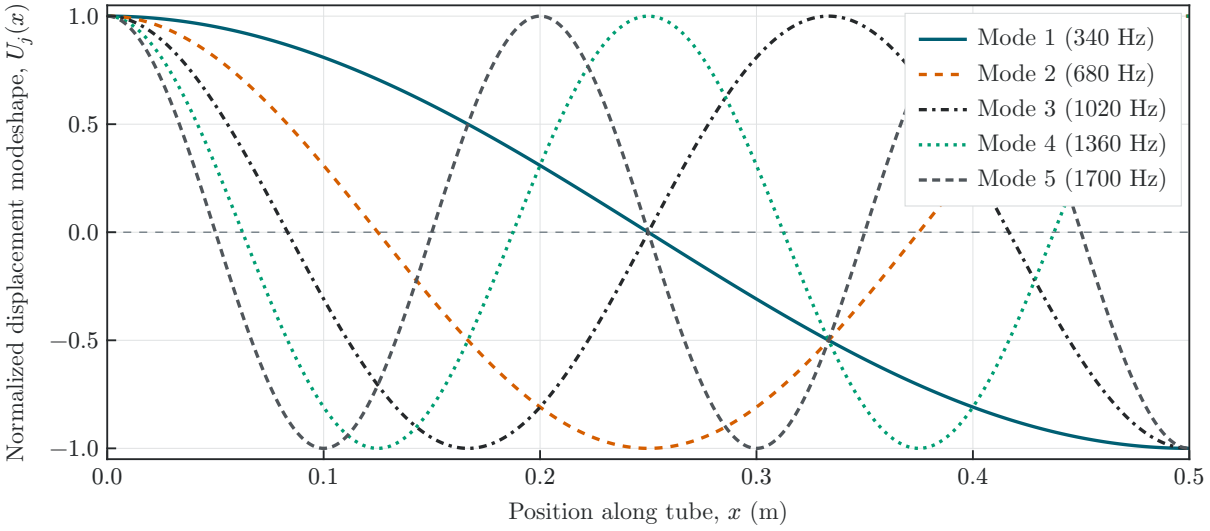

Mode 1 (340 Hz)
Mode 2 (680 Hz)
Mode 3 (1020 Hz)
Mode 4 (1360 Hz)
Mode 5 (1700 Hz)
Normalized displacement modeshape, $U_j(x)$
Position along tube, x (m)
1.0
0.5
0.0
−0.5
−1.0
0.0
0.1
0.2
0.3
0.4
0.5